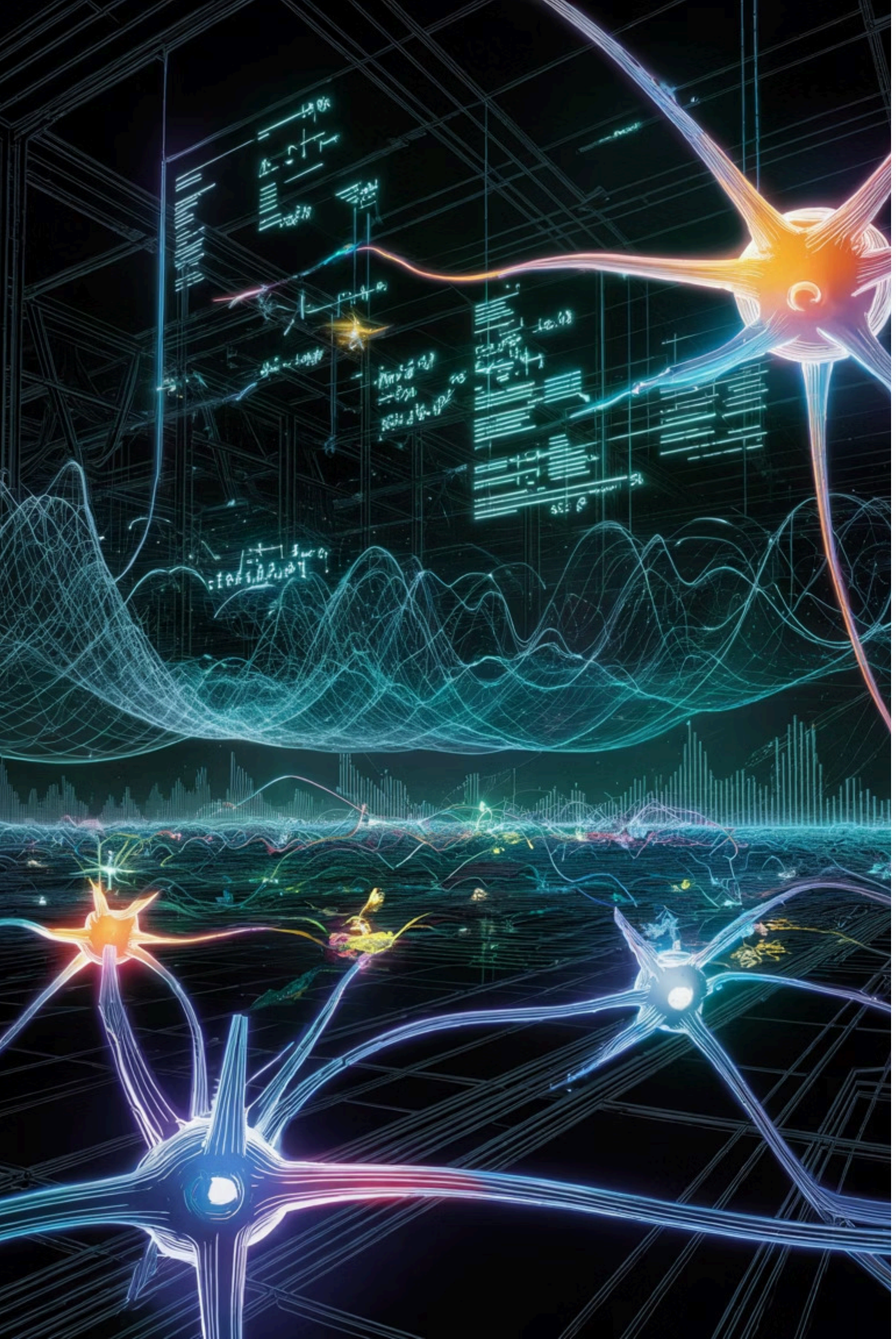

Probability-Based Classification & Advanced ML Concepts

A comprehensive exploration of probabilistic reasoning in machine learning — from foundational probability concepts to Naïve Bayes classification, Association Rule Mining, and Dimensionality Reduction techniques.

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Revision of Probability Concepts

Before diving into classification algorithms, it is essential to revisit the fundamental probability concepts that underpin all probabilistic machine learning methods. Probability provides the mathematical language for quantifying uncertainty — a core challenge in data-driven decision-making.

Core Definitions

Experiment: Any process with an uncertain outcome (e.g., rolling a die).

Sample Space (S): The set of all possible outcomes of an experiment.

Event (E): A subset of the sample space — one or more outcomes.

Probability P(E): The likelihood of event E occurring, where $0 \leq P(E) \leq 1$.

Key Probability Rules

- **Classical Probability:** $P(E) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$
- **Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Multiplication Rule:** $P(A \cap B) = P(A) \cdot P(B|A)$
- **Complement:** $P(\neg A) = 1 - P(A)$
- **Conditional Probability:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$

These rules form the backbone of probabilistic classifiers and are applied repeatedly in Naïve Bayes and other ML algorithms.

Prior Probability

Prior Probability, denoted as $P(H)$, represents the initial degree of belief in a hypothesis **before** observing any new evidence. It is the baseline probability derived from historical data, domain knowledge, or assumptions — independent of the current observation.

Definition

Prior probability is the probability assigned to an event or hypothesis before incorporating any new data or evidence. It reflects pre-existing knowledge or assumptions about the distribution of classes in the dataset.

Example in Classification

If 70% of emails in a dataset are "Not Spam" and 30% are "Spam," then:

P(Spam) = 0.30 (Prior)

P(Not Spam) = 0.70 (Prior)

These priors are used as the starting point before examining the content of any specific email.

Types of Priors

Informative Prior: Based on existing knowledge or previous studies.

Non-informative (Uniform) Prior: Assumes all outcomes are equally likely when no prior knowledge exists.

Empirical Prior: Estimated directly from training data frequencies.

Posterior Probability & Bayes' Theorem

Posterior Probability, denoted as $P(H|E)$, is the updated probability of a hypothesis **after** observing evidence E. It combines the prior belief with new data using **Bayes' Theorem**, making it the cornerstone of probabilistic classification.

Bayes' Theorem Formula

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Where:

- **P(H|E)** = Posterior (probability of H given evidence E)
- **P(E|H)** = Likelihood (probability of E given H)
- **P(H)** = Prior (probability of H before evidence)
- **P(E)** = Marginal likelihood / Evidence

Prior vs. Posterior

Prior: What we believe *before* seeing data.
Likelihood: How probable the data is under each hypothesis.
Posterior: What we believe *after* updating with data.

The posterior is proportional to: **Likelihood × Prior**. The denominator P(E) acts as a normalizing constant ensuring all posteriors sum to 1.

In classification, we compute the posterior for each class and assign the observation to the class with the **highest posterior probability** — this is the Bayes Decision Rule.

Introduction to Naïve Bayes Classification

Naïve Bayes is a family of probabilistic classifiers based on applying Bayes' Theorem with a strong (naïve) independence assumption between features. Despite its simplicity, it performs remarkably well in many real-world applications, especially text classification, spam filtering, and sentiment analysis.

Why "Naïve"?

The algorithm assumes that all features are **conditionally independent** given the class label. This is rarely true in practice, but the simplification makes computation tractable and often yields excellent results.

How It Works

For each class, compute the posterior probability using the product of individual feature likelihoods. The class with the **maximum posterior** is selected as the prediction.

Applications

Spam detection, document classification, sentiment analysis, medical diagnosis, and recommendation systems are among the most common real-world uses of Naïve Bayes.

Working of the Naïve Bayes Algorithm

The Naïve Bayes algorithm follows a structured, step-by-step process to classify new observations. Understanding each step clarifies how prior knowledge and observed evidence combine to produce a classification decision.

The diagram above illustrates the complete classification pipeline — from data preparation through posterior computation to the final class assignment.

Step-by-Step Breakdown

- **Step 1 — Training:** Estimate prior probabilities P(C) for each class from the training data.
- **Step 2 — Likelihood:** For each feature, compute $P(X_i|C)$ — the probability of that feature value given each class.
- **Step 3 — Independence Assumption:** Multiply all individual likelihoods: $P(X|C) = \prod P(X_i|C)$
- **Step 4 — Posterior:** Apply Bayes' Theorem: $P(C|X) \propto P(C) \times \prod P(X_i|C)$
- **Step 5 — Prediction:** Assign the class C that maximizes P(C|X) — the **Maximum A Posteriori (MAP)** estimate.

Key Formula (Naïve Form)

$$P(C|X) \propto P(C) \prod_{i=1}^{n} P(X_i|C)$$

Where **$X = (X_1, X_2, ..., X_n)$** is the feature vector and **C** is the class label. The denominator P(X) is omitted because it is constant across all classes.

Naïve Bayes: Numerical Example

A concrete numerical example demonstrates how Naïve Bayes computes posteriors and makes classification decisions. Consider a simple weather dataset for predicting whether to "Play Tennis."

Outlook	Temperature	Humidity	Play?
Sunny	Hot	High	No
Sunny	Hot	Normal	No
Overcast	Hot	High	Yes
Rain	Mild	High	Yes
Rain	Cool	Normal	Yes

Given: Outlook=Sunny, Temp=Cool, Humidity=High

Priors:

P(Yes) = 3/5 = 0.6

P(No) = 2/5 = 0.4

Likelihoods for "Yes":

P(Sunny | Yes) = 1/3

P(Cool | Yes) = 1/3

P(High | Yes) = 1/3

Likelihoods for "No":

P(Sunny | No) = 2/2 = 1

P(Cool | No) = 0/2 = 0

P(High | No) = 1/2

Posterior Scores

P(Yes | X) ∝ 0.6 × (1/3) × (1/3) × (1/3) = **0.0222**

P(No | X) ∝ 0.4 × 1 × 0 × (1/2) = **0**

Since P(Yes | X) > P(No | X), the prediction is **Play = Yes**.

⚠ Note: P(Cool | No) = 0 causes the entire posterior for "No" to become zero. In practice, **Laplace Smoothing** (adding 1 to all counts) is used to avoid zero probabilities.

Email Classification Using Bernoulli Distribution

The **Bernoulli Naïve Bayes** model is specifically designed for binary/boolean features — making it ideal for text classification tasks like spam detection, where features represent the **presence or absence** of a word in a document.

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Bernoulli Model

Each feature X_i is binary: $X_i \in \{0, 1\}$, where 1 = word present, 0 = word absent. The model explicitly considers both the presence *and* absence of terms, unlike Multinomial Naïve Bayes which only counts occurrences.

Email Classification Setup

Classes: $C \in$ {Spam, Not Spam}. Features: vocabulary words (e.g., "free", "winner", "meeting"). Each email is represented as a binary vector indicating which words appear. Priors and likelihoods are estimated from the labeled training corpus.

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Bernoulli Likelihood

$P(X_i|C) = p_{ic}^{X_i} \cdot (1 - p_{ic})^{(1-X_i)}$

Where p_{ic} is the probability of word i appearing in class C. If X_i=1, we use p_{ic}; if X_i=0, we use $(1 - p_{ic})$ — both contribute to the posterior.

Association Rule Mining – Market Basket Analysis

Association Rule Mining discovers interesting relationships (patterns) between variables in large datasets. **Market Basket Analysis** is its most famous application — identifying which products are frequently purchased together to inform cross-selling, promotions, and store layout decisions.

Key Metrics

Support: How frequently an itemset appears.

$\text{Support}(A \Rightarrow B) = \frac{\text{count}(A \cup B)}{N}$

Confidence: How often the rule is true.

$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$

Lift: How much more often A and B occur together vs. independently.

$\text{Lift}(A \Rightarrow B) = \frac{\text{Confidence}(A \Rightarrow B)}{\text{Support}(B)}$

Lift > 1 indicates positive association; Lift = 1 means independence.

Apriori Algorithm

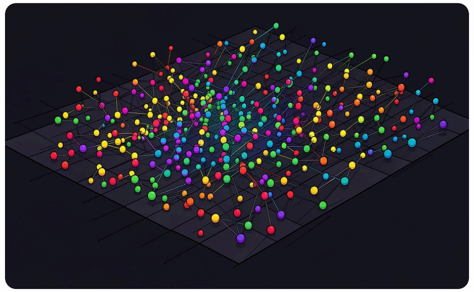
The most widely used algorithm for association rule mining. It uses the **Apriori Principle**: if an itemset is frequent, all its subsets must also be frequent. This allows pruning of the search space.

- Scan database to find frequent 1-itemsets
- Generate candidate (k+1)-itemsets from frequent k-itemsets
- Prune candidates that have infrequent subsets
- Repeat until no more frequent itemsets exist
- Generate rules from frequent itemsets meeting min confidence

Example Rule: {Diapers} ⇒ {Beer} with Support=2%, Confidence=60%, Lift=3.5 — customers buying diapers are 3.5× more likely to buy beer.

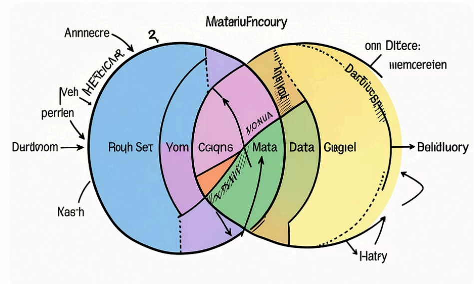
Dimensionality Reduction – PCA & Reduct

Dimensionality Reduction transforms high-dimensional data into a lower-dimensional space while preserving as much meaningful information as possible. It addresses the **Curse of Dimensionality**, reduces noise, improves visualization, and speeds up ML algorithms.



Principal Component Analysis (PCA)

A **linear, unsupervised** technique that finds orthogonal directions (principal components) of maximum variance in the data. The first PC captures the most variance; each subsequent PC captures the next highest, orthogonal to previous ones. Data is projected onto the top-k components, reducing dimensions while retaining maximal information. PCA uses **eigenvalue decomposition** of the covariance matrix.



Reduct (Rough Set Theory)

A **reduct** is a minimal subset of attributes that preserves the classification power of the full attribute set. Based on **Rough Set Theory** (Pawlak, 1982), it identifies the smallest feature subset that maintains the same indiscernibility relation and decision boundaries. Unlike PCA, reducts retain original interpretable features — making them valuable in domains requiring explainability (e.g., medicine, finance).

Key Takeaway: This week covered the full probabilistic ML pipeline — from Bayes' Theorem and Naïve Bayes classification (with Bernoulli models for text) to pattern discovery via Association Rules and feature compression via PCA and Reducts. These tools form the foundation of modern data mining and intelligent systems.